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QUARTERLY EXAMINATION - 2026

12 - STD

PHYSICS

Time : 3.00 Hrs **Marks: 70**

Part – I

Note: 1) Answer all the questions. **15 x 1 = 15**

2) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. An electric field $\vec{E} = 10\,x\hat{\imath}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at x = 2 m is
   a) 10 V b) -20 V c) +20 V d) -10 V
2. A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
   a) Capacitance b) Charge c) Voltage d) Energy density
3. Work done in moving 200μC charge between two points on an equipotential surface is
   a) zero b) finite positive c) finite negative d) infinite
4. The resistance of wire is 20Ω. The wire is stretched to three times its length, then the resistance will now be
   a) 20Ω b) 60Ω c) 120Ω d) 180Ω
5. In a large building, there are 15 bulbs of 40W, 5 bulbs of 100W, 5 fans of 80W and 1 heater of 1kW are connected. The voltage of electric mains is 220V. The minimum capacity of the main fuse of the building will be
   a) 14A b) 8 A c) 10 A d) 12 A
6. A circular coil of radius 5cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is
   a) 1.0Am^2 b) 1.2 Am^2 c) 0.5 Am^2 d) 0.8 Am^2
7. At what distance from a long straight wire carrying a current of 12A will the magnetic field be equal to 3×10^{-5} T?
   a) 8cm b) 12cm c) 18cm d) 24cm
8. The instantaneous values of alternating current and voltage in a circuit are $i = \frac{1}{\sqrt{2}}\sin(100\pi t)\ A$ and $v = \frac{1}{\sqrt{2}}\sin\left(100\,\pi t + \frac{\pi}{3}\right) V$. The average power in watts consumed in the circuit is
   a) $\frac{1}{4}$ b) $\frac{\sqrt{3}}{4}$ c) $\frac{1}{2}$ d) $\frac{1}{8}$
9. $\frac{20}{\pi^2}\ H$ inductor is connected to a capacitor of capacitance C. The value of C in order to impart maximum power at 50 Hz is
   a) 50 μF b) 0.5 μF c) 500 μF d) 5 μF
10. A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux (Φ_B) in weber linked with the coil varies with time (t) second as $\Phi_B = 50t^2+4$, the current in the coil at t=2s is
    a) 0.5 A b) 0.1 A c) 2 A d) 1 A
11. The magnetic field in a electromagnetic wave travelling in vaccum has a peak value of 20nT. The peak value of electric field is a) 9 V/m b) 12 V/m c) 3 V/m d) 6 V/m
12. Which of the following is an electromagnetic wave?
    a) α-rays b) β-rays c) γ-rays d) all of them
13. A rod of length 10cm lies along the principal axis of a concave mirror of focal length 10cm in such a way that its end closer to the pole is 20cm away from the mirror. The length of the image is
    a) 2.5cm b) 5 cm c) 10 cm d) 15 cm
14. Stars twinkle due to
    a) reflection b) total internal reflection c) refraction d) polarisation
15. If red light in Young's double slit experiment is replaced by monochromatic blue light of same intensity, fringe width will
    a) decrease b) increase c) remain uncharged d) will become less intense

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Part – II

Note: Answer any SIX questions. Question No 24 is compulsory. 6 x 2 = 12

16. What is corona discharge?
17. Calculate the electric flux through the rectangle of side 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1}. The angle θis 60°.
18. Define electrical resistivity.
19. State Ampere's circuital law.
20. Fleming's right-hand rule.
21. What is interference of light?
22. What is displacement current?
23. Why does sky appear blue?
24. The self inductance of an air core solenoid is 4.8mH. If its core is replaced by iron core, then its self inductance becomes 1.8H. Find out the relative permeability of iron.

Part – III

Note: Answer any SIX questions. Question No. 33 is compulsory. 6 x 3 = 18

25. Obtain the expression for capacitance for a parallel plate capacitor.
26. State Kirchhoff's laws.
27. A copper wire of cross-sectional area 0.5mm^2 carries a current of 0.2A. If the free electron density of copper is 8.4x10^{28} m^{-3} then compute the drift velocity of free electrons.
28. How do you convert a galvanometer into a voltmeter? Explain.
29. Mention the various energy losses in a transformer.
30. What is meant by Fraunhofer lines? How are they useful in the identification of elements present in Sun?
31. Derive the relation between f and R for a spherical mirror.
32. State Huygens' principle?
33. Light travels from air in to glass slab of thickness 50 cm and refractive index 1.5
    i) What is the speed of light in glass?
    ii) What is the time taken by the light to travel through the glass slab?
    iii) What is the optical path of the glass slab?

Part – IV

Note : Answer all the questions. 5 x 5 = 25

34. a) i) Define electrostatic potential.
    ii) Derive an expression for electrostatic potential due to point charge.

    (or)

    b) Describe the Fizeau's method to determine the speed of light.
35. a) Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle.

    (or)

    b) Obtain an expression for electric field due to dipole on its axial line.
36. a) Obtain an relation for the magnetic induction at a point along the axis of a circular coil carrying current.

    (or)

    b) Obtain the equation for band width in Young's double slit experiment.
37. a) i) Write any six properties of electromagnetic waves.
    ii) If the relative permeability and relative permittivity of the medium are 1.0 and 2.25 respectively. Find the speed of the electromagnetic wave in this medium. **(or)**

    b) i) Define Q factor
    ii) A 500 μH inductor, $\frac{80}{\pi^2}$ PF capacitor and a 628Ω resistor are connected to form a series RLC circuit. Calculate the resonant frequency and Q factor of this circuit at resonance.
38. a) Obtain the condition for bridge balance in Wheatstone's bridge. **(or)**

    b) Obtain lens maker's formula and mention its significance.

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